When does a river forecast change a decision?

Flood warning at Gandak Barrage and water allocation on the Potomac

Abstract

We tested when river forecasts help people issue flood warnings or decide how much water to promise. At Gandak Barrage in India, upstream water levels and archived rainfall forecasts gave six hours of warning before the one qualifying high-flow event in 2026. The same model overestimated the subsequent crest by 7,384 m³/s. Across 809 forecasts, its average crest error was 4,715 m³/s when the upstream gauge exceeded its training range and 1,673 m³/s within that range. Earlier monsoons lacked the hourly barrage records needed to validate this warning model. A separate daily model trained in 2024 missed the one assessable crossing in 2025. On the Potomac River, we replayed water commitments against municipal diversions reconstructed from USGS records on 2,041 days. Planning with the lower end of the flow forecast cut overcommitment from 35.7% of days to 0.9%, while leaving an additional 100,674 m³/day of demand unmet on average. Under the simulated allocation rule, the cost of promising water that could not be supplied had to exceed about four times the cost of leaving the same volume unmet for conservative planning to pay off.